Large language and multimodal models for automatic dietary assessment: Architectural evolution, current challenges, and future perspectives

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ABSTRACT

Background: Accurate dietary assessment is the cornerstone of nutritional epidemiology and precision health. Traditional memory-based methods, such as the 24-hour dietary recall (24HR) and Food Frequency Questionnaires (FFQ), are heavily compromised by systematic biases and participant cognitive burdens. While previous computer vision technologies successfully automated basic food recognition, they struggled with the complex semantic and volumetric reasoning required for precise nutrient quantification. The recent advent of Large Language Models (LLMs) and Large Multimodal Models (LMMs) represents a discontinuous innovation, functioning not merely as direct classifiers, but as semantic reasoning engines capable of deep contextual understanding.

Scope and approach: This review distinctly dissects the architectural evolution of LLM-driven dietary assessment systems from 2023 to 2026. We delineate the transition from the exploratory zero-shot applications of generalist models to the critical development of Phase II methodological enhancements. Specifically, we examine how the integration of Retrieval-Augmented Generation (RAG), Alignment and Fine-tuning strategies, and Chain-of-Thought (CoT) reasoning repairs inherent model deficits. Furthermore, the review explores the frontier of Phase III autonomous agentic systems, detailing the implementation of Mixture-of-Experts (MoE) frameworks, closed-loop continuous monitoring, and external tool utilization.

Key findings and conclusions: While unconstrained generalist models exhibit severe cultural bias and a tendency toward confident hallucination , the application of engineered pipelines—such as RAG and CoT—effectively transforms LLMs into rigorous, evidence-based analytical instruments. Advanced agentic architectures successfully tackle the complexity of real-world meals by dynamically routing tasks to specialized expert modules. To achieve true clinical-grade, automated precision nutrition, future research must overcome the physical bottlenecks of monocular volume estimation, expand ontological adaptability for non-Western diets, and resolve the privacy paradox through Edge AI deployment.

1. Introduction

1.1. Limitations of traditional dietary assessment methods

Accurate dietary assessment constitutes the cornerstone of nutritional epidemiology and precision health, serving as the fundamental prerequisite for establishing causal links between diet and chronic disease outcomes. However, the field has long struggled with inherent methodological limitations that compromise the accuracy of individual intake estimation. For decades, the gold standards in nutritional surveillance—namely the 24-hour dietary recall (24HR) and Food Frequency Questionnaires (FFQ)—have relied predominantly on self-reported data. As comprehensively delineated in foundational epidemiological reviews by Naska et al. (2017) and Bailey (2021), these memory-based instruments are inherently compromised by systematic biases, including recall decay, social desirability bias, and the difficulty of accurately estimating portion sizes without reference standards. The cognitive burden placed on participants often leads to significant underreporting of energy intake, particularly concerning "guilty" foods high in fat or sugar, thereby introducing non-random errors that skew public health data and personalized dietary interventions(Bailey, 2021).

To mitigate the fallibility of human memory, the past decade witnessed a surge in technology-assisted solutions, specifically the application of Machine Learning (ML) and Deep Learning (DL) to food image analysis. Comprehensive surveys on food computing, such as those by Min et al. (2019a) and Wang et al. (2022), highlighted the transformative potential of Convolutional Neural Networks (CNNs) in automating food recognition. These systems achieved remarkable proficiency in classification tasks—correctly identifying a pixel region as "apple" or "pizza"—and demonstrated

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that automated visual analysis could theoretically surpass manual coding in objectivity. Furthermore, Zhu et al. (2021) and Zhang et al. (2023) in their respective surveys on deep learning for food processing and category recognition, emphasized that machine vision technologies have matured sufficiently to handle complex industrial sorting and grading tasks.

The architectural evolution of deep learning in dietary assessment progressed through several pivotal stages. Early works predominantly employed CNN architectures derived from ImageNet pretraining, including ResNet (He et al. (2016)), EfficientNet (Tan and Le (2019)), and MobileNetV2 (Sandler et al. (2018)), transferring learned visual representations to food domain adaptation. To address the challenge of distinguishing multiple overlapping food items within a single plate, attention mechanisms were integrated into encoder-decoder architectures. Research by Min et al. (2020) demonstrated that stacked global-local attention networks significantly improved recognition accuracy on large-scale food datasets. Similarly, Min et al. (2019b) proposed an ingredient-guided cascaded multi-attention network (IG-CMAN) that generates attentional regions from category-supervised networks with a Spatial Transformer, combined with LSTM to discover diverse attentional regions at fine-grained ingredient levels sequentially.

The advent of Vision Transformers (ViTs) marked a paradigm shift in food image analysis, providing effective solutions for fine-grained food classification with similar appearances but distinct nutritional profiles, Gao et al. (2024). Hybrid CNN-Transformer models further enhanced fine-grained food recognition by combining CNNs' local texture capture with transformers' global context modeling, Xiao et al. (2025). Beyond visual feature extraction, knowledge-driven approaches emerged as a promising direction. Knowledge graphs (KGs) model culinary relationships between ingredients and nutritional attributes Chen et al. (2020), while Graph Neural Networks (GNNs) capture higher-order dependencies in food-item relationships Song et al. (2023). The end-to-end paradigm reached a significant milestone with vision-based nutrient estimation pipelines that directly map raw image pixels to nutritional outputs, Moumane et al. (2023); Romero-Tapiador et al. (2025b), and Kwan et al. (2025) introduced NuNet for nutrition estimation utilizing depth sensing.

Nevertheless, a clear distinction persists between identifying food and accurately assessing its nutritional content. While CV models excel at surface-level classification, they encounter substantial obstacles in the semantic and volumetric reasoning required for precise nutrient quantification. This gap is primarily driven by three methodological constraints. First, traditional machine learning approaches are frequently tailored to narrow tasks and small, homogeneous datasets—often restricted to specific populations or predefined food lists—resulting in a sharp decline in performance when applied to cross-cultural cuisines, non-standard dietary presentations, or complex mixed dishes (Oliveira Chaves et al., 2023; Gilal et al., 2024; Zheng et al., 2024; Phalle and Gokhale, 2025). Second, the architectural rigidity of these systems means that incorporating additional food categories or nutrient metrics typically necessitates labor-intensive data re-annotation and model retraining, making it structurally difficult to scale from simple calorie counting to high-dimensional nutrient profiles (Lo et al., 2020; Zheng et al., 2024). Finally, in the specific domain of image-based quantification, traditional visual models remain highly sensitive to environmental variables such as lighting conditions, camera angles, and container scaling; they often rely on specialized geometric modules or depth estimation that lack robustness, struggling to generalize from controlled settings to the ambiguity of real-world eating scenarios (Lo et al., 2020; Shonkoff et al., 2023). Thus,context-aware quantification remains an unresolved frontier.

1.2. Emergence of large language and multimodal models

The advent of Large Language Models (LLMs) and their visual counterparts, Large Multimodal Models (LMMs), represents a discontinuous innovation in food science, fundamentally distinct from the previous generation of CNN-based classifiers. Unlike traditional deep learning models that map pixels directly to labels, LMMs function as Semantic Reasoning Engines. They possess the emergent capability to bridge the gap between low-level visual features and high-level nutritional knowledge graphs.

By pre-training on vast corpora of text and images, models such as GPT-4V and Gemini have acquired an implicit understanding of culinary logic and physical properties. This allows them to perform deductive reasoning that was previously impossible: identifying that a "shiny" texture on grilled vegetables likely implies the presence of oil, or inferring the probable ingredients of a complex stew based on regional cuisine patterns, even when those ingredients are occluded. As discussed in recent literature regarding the application of LLMs in food science (Ma et al., 2024), these models do not merely "see" food; they "understand" the context of consumption. They act as a cognitive interface that can translate a user's vague natural language description (e.g., "a handful of nuts") or a messy meal photo into structured nutritional data by querying internal knowledge bases or retrieving external data. This shift from pattern recognition to reasoning enables the integration of contextual metadata—such as time of day or restaurant type—into the assessment process, theoretically bypassing the limitations of "hidden ingredients" through logical inference.

1.3. Scope and contribution

Despite the rapid proliferation of research at the intersection of AI and nutrition, the current landscape of academic reviews has largely failed to capture the architectural nuances of this fast-evolving field. Existing reviews published in 2025 have predominantly focused on broad clinical applications or generic performance benchmarks. For instance, Tian et al. (2025) provided a comprehensive overview of recognition tasks but remained tethered to static identification paradigms. Similarly, Belkhouribchia and Pen (2025) offered a high-level panorama of LLMs in clinical settings, and Bergling et al. (2025) focused specifically on recommendation systems, leaving the specific technological trajectory of assessment accuracy underexplored. Early investigations like Fazel (2024) primarily tested the zero-shot accuracy of foundation models like GPT-3.5/4, a methodology that is already becoming obsolete as the field moves towards compound systems.

This review fills a critical gap in the literature by focusing distinctively on the architectural evolution of dietary assessment systems between 2023 and 2026. We argue that the field is currently undergoing a second paradigm shift: moving beyond simple Retrieval-Augmented Generation (RAG)—where models merely look up nutritional values—to Autonomous Agentic Systems. This article uniquely dissects how recent innovations, such as Chain-of-Thought (CoT) reasoning, Mixture-of-Experts (MoE) frameworks, and closed-loop multi-agent collaboration, are solving the persistent problems of hallucination and volume estimation. By synthesizing evidence from cutting-edge studies, we delineate the trajectory from static, single-turn analysis to dynamic, interactive, and self-correcting nutritional intelligence, providing a roadmap for the next generation of dietary assessment tools.

2. Zero-shot capabilities of generalist models

The integration of Large Language Models (LLMs) into nutritional science initially focused on leveraging the "zero-shot" reasoning capabilities of foundation models—that is, their ability to perform complex analytical tasks without task-specific training or fine-tuning. Before the advent of specialized agents, the field witnessed a proliferation of exploratory studies aimed at determining whether generalist models, such as GPT-4, Claude, and Gemini, could serve as immediate substitutes for human dietitians or traditional coding systems. This phase of research was characterized by a rigorous examination of two distinct modalities: the semantic parsing of textual dietary recalls and the direct visual estimation of food images.

2.1. Text-based dietary estimation and interaction

For decades, the analysis of 24-hour dietary recalls (24HR) has been a bottleneck in nutritional epidemiology, heavily reliant on the manual mapping of natural language descriptions (e.g., "a bowl of chicken soup") to rigid food composition codes. This process is not only labor-intensive but also prone to inter-coder variability. A pivotal advancement in addressing this inefficiency was demonstrated by Carrillo-Larco (2025). This study provided the first robust evidence that LLMs, when prompted with Chain-of-Thought (CoT) protocols, could effectively bypass the traditional coding pipeline. By utilizing a 10-shot prompting strategy on National Health and Nutrition Examination Survey (NHANES) data, the authors showed that a parameter-efficient fine-tuned model could estimate energy and macronutrients directly from unstructured text strings with a precision comparable to standard databases.

The technical significance of Carrillo-Larco's findings lies in the demonstration of the model's latent semantic alignment. The study revealed that even without accessing an external database in real-time, the model's internal weights—calibrated on vast corpora of nutritional literature—could perform a probabilistic "lookup" function. Specifically, when the model was guided to decompose a complex meal description (e.g., "turkey sandwich with mayo") into its constituent ingredients before estimating values, the Pearson correlation coefficient for macronutrient estimation improved significantly compared to direct inference. This suggests a transformative potential for epidemiological research, implying that the semantic reasoning engine of an LLM can internally "look up" and "compute" nutritional values by leveraging its vast pre-training on food literature, thereby democratizing dietary assessment in resource-constrained settings where manual coding expertise is scarce. Beyond mere quantification, the semantic capabilities of LLMs have been extensively tested in the domain of clinical dietary counseling, where the accuracy of advice is a matter of patient safety. The application of conversational agents in managing chronic diseases represents a shift from static databases to interactive support systems. In the context of renal disease, where the management of electrolytes like potassium and phosphorus is critical, Qarajeh et al. (2023) conducted a comparative analysis evaluating ChatGPT, Bard AI, and Bing Chat. Their findings revealed a disturbing paradox: while these models exhibited a high capacity for identifying general renal-friendly foods (e.g., correctly recommending apples over bananas), they struggled with the nuance of "hidden" mineral content in complex meals. The models frequently failed to account for additive phosphorus in processed foods or the potassium leaching effects of specific cooking methods (like boiling vegetables). This "surface-level competence" raises serious concerns about the consistency of safety-critical advice, as the models often hallucinated safe consumption limits that contradicted established renal care guidelines.

This inconsistency was further echoed in the work of Hoang et al. (2023), who assessed the performance of AI across English and Traditional Chinese inputs. They observed a significant "linguistic capability gap." While the energy estimation for single, standardized food items (e.g., a medium-sized apple) was generally accurate and consistent across languages, the advice for composite meals varied significantly depending on the language used. The study highlighted that when queried in Traditional Chinese, the models were more likely to hallucinate nutritional values for Western-style dishes, or conversely, fail to recognize the caloric density of traditional Asian cooking methods (such as braising with sugar). This highlights a linguistic vulnerability in global deployment, suggesting that the model's "nutritional knowledge" is not a language-agnostic concept but is deeply tethered to the linguistic frequency of training data.

The complexity increases further when addressing metabolic syndromes. Naja et al. (2024) explored this by benchmarking ChatGPT's outputs against international clinical guidelines for diabetes and metabolic syndrome management. The study underscored that while LLMs could generate "plausible" meal plans that seemingly adhered to caloric restrictions (e.g., a 1500 kcal/day plan), they often lacked the clinical precision required to tailor macronutrient distribution to an individual's glycemic response. For instance, the generated meal plans frequently failed to distinguish between high-glycemic and low-glycemic index carbohydrates, recommending fruit juices or refined grains that, while fitting the calorie budget, would be deleterious for blood glucose control.

Similarly, in the specialized field of urolithiasis, Aiumtrakul et al. (2024) investigated the utility of chatbots for preventing kidney stones. Their results indicated that while the model could successfully categorize high-oxalate foods, it failed to account for bioavailability factors—such as the calcium-oxalate interaction during digestion. A human dietitian understands that consuming calcium-rich foods alongside oxalate-rich foods can reduce stone formation risk by binding oxalate in the gut; however, the LLM, lacking a causal model of human physiology, often provided simplistic advice to merely "avoid oxalates," missing the nuanced dietary synergy that is central to modern medical nutrition therapy. Collectively, these studies from 2023-2025 demonstrate that while text-based LLMs have achieved a "competent amateur" status in nutritional coding and general advice, they largely lack the "clinical intuition" required for managing complex pathologies without external knowledge augmentation.

2.2. Image-based direct estimation

The emergence of Large Multimodal Models (LMMs) such as GPT-4V (Vision), Gemini 1.5 Pro, and Claude 3.5 Sonnet sparked a wave of enthusiasm for "image-based direct estimation"—the holy grail of automated dietary assessment. Unlike previous computer vision (CV) methods that required separate detection and volume estimation modules, these end-to-end models promise to extract nutritional data directly from a raw pixel input.

A landmark comparative study by Fridolfsson et al. (2025) subjected GPT-4o, Claude 3.5 Sonnet, and Gemini 1.5 Pro to a rigorous test using standardized food photography. The results exposed a critical systematic bias that can be described as the "Volumetric-Density Paradox." While the models demonstrated superior food recognition capabilities compared to legacy CNNs (e.g., correctly distinguishing between different types of pasta shapes), they consistently exhibited poor volumetric reasoning. Specifically, the study found a non-linear error distribution: the models tended to underestimate portion sizes for high-density, amorphous foods (like mashed potatoes, dense curries, or nut butters) while overestimating them for voluminous, low-calorie items (like leafy salads or popcorn).

Fridolfsson's analysis suggests that LMMs rely heavily on 2D segmentation area as a proxy for mass, failing to account for the "thickness" or specific gravity of the food item. Among the three models evaluated, GPT-4o showed a slight edge in reasoning through complex plate compositions, utilizing contextual clues (like plate rim size) better than its competitors. However, all models struggled to achieve the accuracy required for clinical utility (typically defined as <10% error rate), with Mean Absolute Errors (MAE) for calorie estimation often exceeding 20-30% for mixed dishes. This highlights a fundamental disconnect between "seeing" an object and "sensing" its mass in 3D space, a limitation inherent to 2D pre-training data where depth information is implicitly lost.

Complementing this, O'Hara et al. (2025) focused specifically on the dominant model in the field, investigating ChatGPT's sensitivity to visual references. Their analysis revealed that ChatGPT's estimations were highly stochastic and sensitive to the presence of reference objects (like a coin or fork). Crucially, the study found that the mere presence of a fiducial marker was insufficient; the model had to be explicitly prompted to use the marker for scale. Without such "attentional guidance," the model frequently ignored the coin, hallucinating an arbitrary scale for the food portion. This "prompt sensitivity" was systematically analyzed Lo et al. (2024), who demonstrated that Chain-of-Thought (CoT) prompting could significantly reduce estimation error. By forcing the model to articulate its recognition process step-by-step (e.g., Step 1: "I see a 10-inch plate"; Step 2: "The rice covers 1/3 of the plate"; Step 3: "Based on standard density..."), the researchers achieved a measurable reduction in energy estimation error. However, even with advanced prompting, the absolute error rates for mixed dishes remained high, indicating that "logical reasoning" cannot fully compensate for "perceptual blindness" regarding depth and density.

The readiness of these models for real-world deployment was questioned in a comprehensive benchmark by Romero-Tapiador et al. (2025a). Introducing the FoodNExTDB database, the authors evaluated six state-of-the-art VLMs (including Moondream and LLaVA). Their conclusion was stark: while "recognition" is largely solved for common Western foods, "assessment" remains fragile. The study introduced the concept of "ingredient hallucination," observing that models often hallucinated ingredients that were visually plausible but culinarily incorrect. For example, based purely on texture, a model might identify mashed potatoes as hummus, or mistake a soy-based sauce for a beef reduction. This semantic confusion leads to catastrophic errors in macronutrient calculation (e.g., substituting carbohydrates for fats). This echoes the earlier findings of Haman et al. (2024), which was one of the first studies to unveil the accuracy gaps of ChatGPT. Haman noted that while the model could pull accurate data from the USDA database for named foods (e.g., "100g of almonds"), its ability to match an image to those database entries was fraught with error, often selecting the wrong varietal or preparation method. Thus, Phase I of the LMM era has proven that "zero-shot" vision is a powerful classification tool but an unreliable measurement instrument.

2.3. Limitations regarding hallucination and cultural bias

The most profound limitation of generalist models lies not in their inability to count pixels, but in their inherent cultural bias and tendency towards "confident hallucination." LLMs are products of their training data, which is overwhelmingly dominated by the Western Internet—recipes, blogs, and databases from North America and Europe. This creates a "Western Lens" through which all global cuisine is interpreted, leading to catastrophic failures when assessing non-Western diets.

This phenomenon was strikingly illustrated by Carrillo-Larco et al. (2025) in their critical evaluation of Peruvian lunch meals. The study served as a crucial "stress test" for model universality, deliberately stepping outside the comfort zone of USDA-centric datasets. When presented with images of typical Peruvian lunches—such as Lomo Saltado (stir-fried beef) or Ají de Gallina (creamy chicken stew)—generalist models frequently misidentified key ingredients or drastically misjudged caloric density. The study identified a mechanism of failure termed "Cooking Method Bias." For instance, models would often fail to recognize the heavy use of oil in local preparation methods, assuming instead a "standard" (Western) sautéing or grilling process. This led to a systematic underestimation of fat content, sometimes by a factor of two.

More alarmingly, Carrillo-Larco observed that the models often hallucinated "healthy" additions that were not present. For example, when viewing a ambiguous green garnish, the model might confidently label it as "kale" or "spinach" (ingredients common in Western health-conscious datasets), whereas in the Peruvian context, it was likely cilantro or a local herb with a completely different nutritional profile. Similarly, the models frequently hallucinated the use of olive oil (a Mediterranean staple) in dishes that traditionally use lard or vegetable oil, reflecting a statistical bias towards the "healthy eating" corpus the models were fine-tuned on.

This issue of cultural bias is not merely an accuracy problem but an equity issue. As discussed in the broader context of comparison studies by Ponzo et al. (2024), the lack of diverse culinary representation in training sets means that AI-driven nutritional tools work best for the populations that need them least (affluent Westerners with standardized diets) and fail the populations with the highest burden of diet-related disease (Global South populations with complex, mixed-dish diets). The "black box" nature of these models exacerbates the problem; when a model sees a bowl of noodles, it cannot explain why it assigned a certain caloric value, making it impossible for a user to diagnose whether it missed the hidden sugar in the broth (common in Southeast Asian cuisine) or misjudged the density of the noodles. This "illusion of competence"—where the model confidently identifies a dish but completely misunderstands its nutritional essence—remains the single biggest barrier to the unassisted deployment of generalist LLMs in global public health.

3. Methodological enhancements for precision assessment

The critical evaluation of generalist models in the preceding section unveiled a fundamental epistemological paradox in the application of Artificial Intelligence to nutritional science: while foundation models possess an unprecedented breadth of semantic knowledge, they suffer from a distinct lack of domain-specific grounding required for precise nutritional quantification. In the context of clinical nutrition, accuracy is not merely a metric of performance but a requirement for safety. Generalist models, by their architectural nature, tend to "guess" nutritional values based on the statistical probability of next-token prediction rather than "calculate" them based on deterministic factual data. This probabilistic nature leads to a "Black Box" phenomenon where the derivation of a caloric estimate is opaque, unverified, and frequently hallucinatory.

To address this existential challenge, the period between 2024 and 2026 witnessed a profound methodological turn within the research community. Researchers ceased to treat Large Language Models (LLMs) as standalone solutions. Instead, a new engineering paradigm emerged, treating LLMs as modular, malleable components within sophisticated, multi-stage computational pipelines. This phase is characterized by the systematic engineering repair of LLMs' inherent defects—specifically their propensity for hallucination, their visual insensitivity to food composition, and their lack of logical planning.

This remediation is achieved primarily through three architectural pillars that define Phase II: the integration of Retrieval-Augmented Generation (RAG) enables the model to search external databases and knowledge graphs for factual information; the application of Alignment and Fine-tuning strategies—including Low-Rank Adaptation (LoRA), hybrid architectures, and visual feature fusion—modifies model structures and weights to improve its direct comprehension of complex inputs; and the implementation of Chain-of-Thought (CoT) prompting guides the model's reasoning, decomposing complex tasks into structured, step-by-step logical processes. Collectively, these enhancements mark the transition from "Zero-Shot Prediction" to "Controlled, Evidence-Based Reasoning."

3.1. Retrieval-augmented generation for database anchoring

This subsection focuses on how to help models better understand the input through external databases. In the context of dietary assessment, the most pernicious failure mode of zero-shot LLMs is "probabilistic hallucination"—the tendency of the model to generate nutritionally plausible but factually incorrect values (e.g., misidentifying low-fat yogurt as full-fat cream based solely on visual appearance). To fundamentally eliminate this risk, the field is taking use of Retrieval-Augmented Generation (RAG) architecture. RAG fundamentally alters the inference trajectory: instead of forcing the model to rely on its static and error-prone internal weights, it compels the model to function as a semantic search engine, querying external, verified, authoritative knowledge bases before generating any output.

The dual-stream architecture. A quintessential implementation of RAG was demonstrated by Yan et al. (2025), who engineered a comprehensive framework to address the limitations of commercial models in handling large-scale food databases. Recognizing that standard Multimodal Large Language Models (MLLMs) struggle to "memorize" the exact macronutrient profiles of thousands of food items without catastrophic forgetting, the authors devised a dual-stream interactive pipeline. In this novel architecture, the role of the Visual Encoder is fundamentally redefined: it is stripped of the responsibility for direct "quantification" (i.e., predicting calories directly from pixels) and instead focuses solely on high-fidelity "feature extraction" and entity recognition.

The workflow begins with the system identifying food entities within the image and converting them into high-dimensional vector embeddings. These vectors serve as semantic queries that are matched against a pre-indexed vector database derived from the Food and Nutrient Database for Dietary Studies (FNDDS) using cosine similarity search. The core innovation of this approach lies in its "Dynamic Context Injection" mechanism. Once the relevant food items are retrieved from the database, the system does not merely output them; it extracts precise, tabular nutritional metadata (e.g., "100g of roasted boneless chicken breast contains 165 kcal, 31g protein, 3.6g fat") and embeds this verified, deterministic context directly into the LLM's prompt window (Context Window).

By adopting this strategy of "Decoupling Recognition and Quantification," the framework effectively transforms the LLM from a "guesser" into a "synthesizer." The model no longer needs to recall the calorie count of chicken breast; it only needs to read the retrieved context and adjust it based on the estimated portion size. Experimental results indicated that this RAG-driven approach significantly reduced the Mean Absolute Error (MAE) for calorie estimation compared to closed-source commercial models that rely on end-to-end prediction. This finding provides compelling evidence that the future of nutritional AI lies not in simply scaling model parameters to trillions of weights, but in establishing deep, architectural connections with existing, rigorous nutritional science infrastructures.

Ontology-driven semantic bridging. The foundational logic for such semantic matching was laid before the generative AI boom, as explored by Rezayi et al. (2022). Although their work utilized a BERT-based architecture rather than modern generative transformers, it established the critical role of Domain Ontology in bridging the linguistic gap between layperson descriptions and scientific databases. General-purpose models often fail to map colloquial food terms (e.g., "spicy tuna roll") to the rigid, hierarchical nomenclature of food composition databases (e.g., "Sushi, tuna, spicy, with mayonnaise").

By integrating the FoodOn ontology—a rigorous semantic hierarchy of food products—and pre-training the model on a massive corpus of agricultural and food science literature, the authors created a model capable of understanding the deep semantic relationships between food concepts. This "Knowledge-Infused" approach allowed the model to navigate the complex taxonomy of food ingredients, effectively bridging the "Semantic Gap." For instance, the model could understand that "whole grain bread" is a subclass of "bread" with specific nutritional properties distinct from "white bread," a distinction often lost in simple keyword matching. The prescience of this work lies in its demonstration that effective RAG is not merely about retrieving documents based on keyword overlap; it requires a semantic bridge capable of understanding hierarchical and ontological relationships. This principle has now been widely adopted by modern LLM systems to handle the ambiguity inherent in natural language dietary recalls, ensuring that user inputs are accurately mapped to the correct nutritional codes.

Structured knowledge graph reasoning. As the demand for personalized nutrition increases, simple flat database retrieval (looking up a single row of data) is no longer sufficient for complex dietary planning that involves multiple constraints and interactions. Mohbat and Zaki (2025) advanced the RAG concept into the realm of Structured Knowledge Graphs (KG). Unlike standard RAG systems that retrieve unstructured text chunks or simple database rows, their proposed system utilizes a Knowledge Graph to map the intricate topological relationships between Users, Recipes, Ingredients, and Nutrients.

In this graph-based architecture, entities are nodes, and relationships (e.g., "contains," "is_suitable_for," "has_nutrient") are edges. When a user presents a complex, multi-constraint dietary requirement—such as "suggest a low-purine, high-protein dinner for a gout patient that excludes dairy"—a simple vector search would likely fail to capture the logical negation and the specific medical constraints. However, the KG-enhanced system performs Multi-hop Reasoning on the graph. It traverses from the "user" node to the "disease" node (gout), identifies the constraint (avoid purines), hops to "ingredient" nodes to filter out "red meat" and "organ meats," and then follows edges to find "tofu" or "legumes" as valid high-protein substitutes. This "Knowledge-Enhanced" approach allows the system to evolve beyond a mere calculator into a "culinary architect" capable of logical deduction and substitution. The integration of KGs ensures that the generated recommendations are not just statistically probable but logically consistent with a complex web of nutritional and medical rules.

Clinical safety and knowledge infusion. Furthermore, the deployment of RAG involves core issues of clinical ethics and safety, particularly when dealing with vulnerable populations or chronic disease management. Abbasian et al. (2024) addressed this critical dimension by investigating the risks of unconstrained LLMs in providing medical advice. They warned that generalist LLMs, lacking specific medical context, often provide generic or even dangerous advice for chronic conditions like diabetes (e.g., suggesting high-glycemic fruits to a patient with uncontrolled blood sugar).

To mitigate this, they proposed a framework that implements a strict "Knowledge Infusion" strategy. This system utilizes the American Diabetes Association (ADA) clinical practice guidelines and the Nutritionix database not merely as reference material, but as impassable "guardrails" for the model's output. Technically, this is achieved by constraining the model's generation space; every piece of advice generated must be cross-referenced and validated against the retrieved medical protocols. If the model attempts to generate a suggestion that conflicts with the ADA guidelines, the Knowledge Infusion mechanism suppresses that output. In their evaluation, this architecture reduced the rate of clinically unsafe recommendations to near zero. This finding establishes RAG as a non-negotiable requirement for clinical nutrition applications: it effectively transforms the LLM from an unconstrained "creative writer" prone to dangerous improvisation into a regulated, evidence-based "medical assistant."

3.2. Visual-language alignment and fine-tuning strategies

This subsection focuses on how to modify the model to improve its ability to understand the input itself instead of through external databases. While RAG addresses the issue of factual accuracy regarding nutritional data, it relies heavily on the accuracy of the initial visual recognition. The retrieval process is only as good as the query generated from the image. If the model fails to visually distinguish between "deep-fried chicken cutlet" and "air-fried chicken cutlet"—two items that look nearly identical but differ vastly in caloric content due to oil absorption—the retrieved database entry will inevitably be incorrect. Phase II research has profoundly recognized that generalist visual encoders (such as CLIP or SigLIP used in GPT-4V) suffer from a severe "Visual Semantic Gap" in the food domain. While these models are excellent at recognizing Objects (e.g., "this is chicken"), they often fail to perceive the Food Matrix—the physical state of the food that indicates its processing method, oil content, moisture level, and texture density. Consequently, research focus has shifted towards Visual-Language Alignment and Parameter-Efficient Fine-Tuning (PEFT) to sensitize models to these subtle visual cues.

Reshaping the alignment apace with low-Rank Adaptation (LoRA). A breakthrough in addressing this visual insensitivity was proposed by Yao et al. (2024). The authors astutely pointed out that the visual cues for calories and macronutrients are often extremely subtle: the slight sheen of oil on a vegetable, the viscosity of a sauce, or the specific browning pattern of a batter. These features are frequently overlooked in general large-scale pre-training, which prioritizes broad object recognition over fine-grained texture analysis.

To recover these lost signals without the prohibitive cost of retraining a foundation model from scratch, the researchers employed Low-Rank Adaptation (LoRA) technology. LoRA works by freezing the pre-trained model weights (W_0) and injecting trainable rank decomposition matrices (A and B) into the self-attention layers of the Transformer architecture, such that the weight update is represented as $\Delta W = BA$. This approach allows the model to adapt to a new domain while modifying fewer than 1% of its total parameters. By constructing a specialized dataset, CalData, containing 330,000 carefully curated image-text pairs that emphasize ingredient-level and texture-level descriptions, the authors enabled the model to learn a specialized "Calorie-aware Subspace." In this aligned feature space, specific visual textures (e.g., "crispy," "oily," "dense") become tightly coupled with specific nutritional descriptors (e.g., "high-fat," "calorie-dense"). Experimental results demonstrated that the LoRA fine-tuned model significantly reduced the error rate in estimating calories for high-density and processed foods. This proves that lightweight, targeted fine-tuning can effectively endow generalist models with the expert capability to "perceive invisible calories" that are typically missed by standard models.

Correcting scene-dominance aias via feature fusion. Going beyond global parameter tuning, Qi et al. (2025) proposed a more refined architectural innovation to address the "Scene-dominance Bias" prevalent in standard LMMs. They identified that when presented with a food image, generalist models tend to focus on the global layout or the most salient object (e.g., the plate, the table setting, or the main protein), while ignoring key local details that disproportionately influence nutritional value (such as a sprinkling of cheese, a dressing on a salad, or the ratio of rice to meat).

To correct this bias, the authors developed a Visual-Ingredient Feature Fusion (VIF^2) mechanism. This is a model-agnostic plug-in module that fundamentally alters how visual information is processed. Instead of feeding the entire image embedding to the LLM, the system explicitly uses an object detection algorithm to crop and extract Local Visual Features of individual ingredients. These local features are then deeply fused with their corresponding Text Embeddings before entering the LLM's inference layer. This mechanism effectively forces the model to reallocate its Attention Mechanism: instead of attending to the "scene," the model is mathematically compelled to attend to specific "ingredients." By explicitly modeling the visual-semantic interaction at the ingredient level, VIF^2 achieved significantly higher fidelity when processing complex Mixed Dishes. This work shifts in dietary assessment methodologies from being "Scene-centric" (analyzing the meal as a whole) to "Ingredient-centric" (analyzing the weighted sum of components).

Hybrid neuro-nymbolic approaches for packaged foods. In the realm of packaged foods, the nature of the visual challenge shifts from texture recognition to Information Extraction and Reading. The proliferation of Ultra-processed Foods (UPFs) presents a unique challenge: the nutritional content is not defined by visual appearance but by the data printed on the Nutrition Facts label. Hakim et al. (2025) addressed this with the NutriGuard system, which employs a hybrid neuro-symbolic architecture.

NutriGuard recognizes that generalist multimodal models often commit Optical Character Recognition (OCR) errors when reading small, dense, or curved text on food packaging. To solve this, the system integrates a dedicated, deterministic OCR module (PaddleOCR or Surya) with a fine-tuned Llama-3.2 model. Unlike traditional "vision-only" methods that try to "guess" the content of a package based on its logo or shape, NutriGuard directly "reads" the Nutrition Facts Label and the Ingredient List. Subsequently, the LLM component does not simply parrot this data; it performs a semantic interpretation combined with the user's personal health profile. For example, it can automatically flag "hidden sugars" (identifying synonyms like "high fructose corn syrup" or "dextrose") for a diabetic user, or highlight allergens. This hybrid mode of "Visual Reading + Semantic Interpretation" effectively solves the "black box" problem in packaged food assessment, combining the precision of symbolic text extraction with the reasoning power of LLMs.

Feature engineering and model fusion for longitudinal tracking. Simultaneously, addressing the need for optimized data representation in continuous monitoring, Mishra et al. (2024) explored the potential of Lightweight Hybrid Models. While massive foundation models (like GPT-4) offer superior generalized reasoning, longitudinal dietary tracking requires precise feature engineering to handle complex behavioral patterns. To achieve this, the authors proposed an architecture combining BERT (for deep semantic understanding of food names) with LSTM (Long Short-Term Memory networks) for processing dietary time-series data.

This hybrid approach exemplifies effective model fusion by leveraging the distinct strengths of specific architectures to optimize the representation of input data. Specifically, it combines BERT's robust comprehension capabilities to generate rich semantic embeddings of food terminology with the LSTM's memory capacity to capture the temporal dependencies of a user's eating habits (e.g., predicting that a user who eats a heavy breakfast might eat a lighter lunch). By enriching the feature space through this synergistic combination, the model achieves highly accurate contextual awareness. Finally, the inherent efficiency of this hybridized architecture also seamlessly addresses the critical requirements for real-time performance, low latency, and privacy in mobile and edge applications.

3.3. Chain-of-thought methods

The third pillar of methodological optimization in aims to address the "Logical Gap" in the assessment process. A fundamental disconnect exists between how deep learning models and human experts approach the task of calorie estimation. A human dietitian never attempts to assign a calorie value to a meal based on a single, holistic glance. Instead, they follow a strict, structured cognitive protocol: Identification (What is this?) → Decomposition (What are the components?) → Volumetric Estimation (How much of each?) → Calculation (Summing the values). In stark contrast, Phase I models attempted to jump directly from the raw pixel input to the final calorie value via a Single Forward Pass. This cognitive "leap" forces the model to perform implicit reasoning that is often fragile, leading to High Variance and unexplainable errors.

Reasoning-driven estimation and self-correction. To bridge this logical gap, Tanabe and Yanai (2025) introduced Chain-of-Thought (CoT) prompting into the domain of multimodal dietary assessment. Based on the hypothesis that "error stems from a lack of logic," the authors designed a Multi-stage Prompting Strategy that enforces a structured reasoning path. Under this framework, the model is explicitly forbidden from directly outputting the final nutritional values. Instead, it must first generate a "Rationale" or a descriptive intermediate representation.

For example, when presented with an image of Ramen, the model is prompted to first describe the scene in detail: "I see a porcelain bowl about 20cm in diameter, containing approximately 200g of noodles, submerged in a rich tonkotsu broth. The topping includes two slices of pork belly (estimated 30g each) and half a soft-boiled egg." Only after completing this "Textual Grounding" step—which serves to verify the model's visual perception—is the model permitted to invoke its internal knowledge or external tools to perform the calculation. Experimental data showed that this "Reasoning-Driven" approach reduced the Root Mean Square Error (RMSE) to a significant level compared to direct estimation. Crucially, the intermediate reasoning text acts as a "Self-correction Mechanism": it allows the user to inspect the model's thought process. If the model misidentifies "tofu" as "cheese" in the rationale phase, the user can correct the premise before the calculation occurs, a feature impossible in "black box" end-to-end models.

The divide-and-conquer strategy via segmentation. Taking this logical decomposition to its architectural extreme is the "Divide-and-Conquer Strategy" proposed by Khlaisamniang et al. (2025). Inspired by the operational standards of clinical nutrition, where mixed meals are always broken down into ingredients for analysis, the authors built a system that fundamentally rejects "holistic estimation."

The system employs advanced Segmentation Models (such as SAM - Segment Anything Model) to physically cut complex meal images into discrete, semantic sub-regions (e.g., isolating the rice area, the curry sauce area, and the chicken area into separate image patches). The LLM is then tasked with analyzing each sub-region independently.

This "Decomposition-based Approach" effectively achieves Complexity Reduction: it transforms one difficult, high-entropy problem (estimating a messy plate) into several simpler, lower-entropy problems (estimating homogenous components). This strategy prevents the Cognitive Overload that is common in LLMs when dealing with multi-object scenarios where attention heads may become distracted. The study provides compelling evidence that the "Architecture of Thought"—the workflow and logic imposed on the model—is just as important as the architecture of the neural network itself. By imposing a nutritionist-inspired Standardized Workflow on the AI system, we can effectively bridge the gap between stochastic text generation and rigorous scientific assessment.

In summary, Phase II marks the maturation of LLMs in food science applications. It represents a move away from the naive application of chatbots towards the construction of engineered intelligent systems. By connecting models to deterministic knowledge networks via RAG, endowing models with acute visual perception via PEFT, and standardizing model reasoning logic via CoT, researchers have successfully transformed the LLM from a passive, hallucination-prone observer into an active, rigorous, and verifiable analytical instrument. These technological advancements have not only significantly improved assessment precision but have also laid a solid foundation for the evolution towards fully autonomous, tool-using agents in Phase III.

4. Multimodal agents and automated assessment systems

If Phase II represented the "instrumental repair" of generalist Large Language Models (LLMs) through prompting and retrieval enhancements, Phase III marks the genesis of an entirely new "Agentic Architecture." In this advanced phase, the research focus shifts fundamentally from single, static predictive models to dynamic, collaboratively modular Autonomous Agentic Systems. The core driver of this profound paradigm shift is the realization within the computational nutrition community that dietary assessment is inherently an unstructured, multi-step reasoning process. A single set of monolithic neural network weights cannot simultaneously master the disparate cognitive tasks of visual identification, volumetric quantification, nutritional reasoning, and longitudinal dietary planning. Consequently, the latest literature demonstrates a necessary evolution in system design towards Mixture of Experts (MoE) topologies, architectural Divide-and-Conquer strategies, and Closed-Loop Control mechanisms, all aiming to achieve truly fully automated, clinical-grade precision nutrition management.

4.1. Mixture of experts and divide-and-conquer strategies

Real-world meal images possess incredibly high semantic density and visual complexity. A single photograph may contain overlapping ingredients, hidden sauces integrated into the food matrix, and beverages served in varying, non-standardized containers. Traditional monolithic models attempt to solve all these issues via a single forward pass through the network, which inevitably leads to "catastrophic forgetting" or severe interference between varying task gradients.

It is imperative at this juncture to explicitly distinguish the "Divide-and-Conquer" strategies defining Phase III from the image decomposition techniques previously discussed in Phase II. In Phase II, decomposition was primarily a cognitive or pre-processing intervention; researchers used standard segmentation tools to crop images, and then fed these isolated image patches sequentially into the same generalist LLM using structured Chain-of-Thought prompts. While effective, this forced a single neural architecture to act as a universal solver, leading to cognitive overload and high latency. In stark contrast, the Phase III Divide-and-Conquer approach is structurally embedded into the neural architecture itself. Instead of changing the prompt, researchers have changed the brain of the model.

To address the limitations of monolithic processing, Zhang et al. (2025) proposed a paradigm-shifting cognitive architecture that completely abandons the expectation that a single model must "rote memorize" all nutritional knowledge. Instead, their research constructs a sophisticated Multi-modal Mixture of Experts (MoE) system. This advanced framework comprises three physically distinct, specialized reasoning modules engineered for specific food science tasks. The first is a Visual Expert, heavily weighted toward convolutional feature extraction, dedicated solely to identifying visually distinct staple foods and quantifying their spatial boundaries. The second is a Retrieval-Augmented Generation (RAG) Expert, fine-tuned specifically for inferring ambiguous components—such as heavily mixed stews, purees, or composite sauces—by querying external culinary databases to deduce probable ingredient ratios based on regional recipes. The third is a Commonsense Expert, trained to handle trace ingredients, garnishes, and the physical constraints of food pairing.

The operational heart of this system is a sophisticated "Central Dispatcher." This dispatcher leverages the overarching semantic understanding capabilities of a foundational LLM not to analyze the food directly, but to dynamically route different Regions of Interest (RoI) in the input image to the most appropriate downstream expert. For instance, when presented with an image containing both a glass of milk and a complex chicken curry, the dispatcher actively segments the request. It routes the visual data of the milk to the Volume Estimation Expert to calculate liquid height against the glass curvature, while simultaneously triggering the RAG Expert to cross-reference the visual texture of the curry against known recipe databases to estimate hidden fat and sodium content. Experimental results from this architectural divergence demonstrate a substantial leap in efficacy; the dynamic routing mechanism significantly mitigates the mean absolute error (MAE) in handling complex mixed dishes when compared to a monolithic GPT-4V baseline. This empirically proves that architecturally decomposing complex nutritional problems into parallel sub-tasks is the definitive key to enhancing assessment robustness in real-world, multi-component meals.

The rigorous mathematical foundation of this modular design is further explored and quantified by Jiao et al. (2024). Their research team identified a critical phenomenon known as "Gradient Conflict" that occurs during the fine-tuning of large multimodal models on diverse food datasets. They pointed out that when a single network is forced to learn both generative tasks (such as writing fluent recipe instructions) and regression tasks (such as predicting exact caloric and macronutrient numerical values), the optimization pathways inherently conflict. Optimizing the textual fluency of recipe generation actively degrades the numerical precision of the nutrition estimation, as the loss functions pull the model's attention weights in opposing directions. To permanently mitigate this mathematical bottleneck, their team introduced a Linear Rectified Mechanism designed to seamlessly coordinate distinct expert modules within the network. This architecture enables the model to automatically learn how to allocate parameter capacity during the backpropagation training phase. It assigns fewer computational resources and shallower network pathways to simple classification tasks (e.g., identifying an apple), while dynamically allocating drastically more "expert" weights and deeper reasoning layers to complex macronutrient estimations (e.g., calculating the protein-to-fat ratio in a marbled steak). The profound success of this approach lies not only in the documented performance gains across standard benchmarks but in its demonstration of what the authors term "Parameter Economics." By allowing different expert networks to specialize and selectively activate, the system achieves a comprehensive, high-fidelity understanding of food's multidimensional attributes—encompassing taste profiles, nutritional density, and implied cooking methods—without significantly increasing the computational overhead or inference latency required for mobile deployment.

Furthermore, the versatility of such expert-driven systems is masterfully showcased by Yin et al. (2025). Their development of a highly versatile, multi-modal food assistant proves that expert integration can extend beyond backend calculations into frontend user interaction. By integrating specialized Segmentation Heads alongside dedicated Nutrition Heads within the model architecture, their system can understand food matrices at the granular pixel level while seamlessly transitioning computational roles during a live user conversation. The model acts with striking fluidity, shifting from a "Classification Expert" that rigorously maps ingredient boundaries on the user's screen, to a "Culinary Consultant" that leverages its generative capacities to suggest lower-sodium alternative recipes based on the recognized ingredients. This unprecedented architectural flexibility strongly suggests that the future of dietary assessment lies in systems that are not single-purpose, rigid analytical tools, but comprehensive, highly adaptive assistants. These future systems will possess a form of "Liquid Intelligence," capable of instantaneously reorganizing their internal expert modules and attention mechanisms based on the immediate, shifting needs of the user—whether that entails executing a rigid caloric calculation or engaging in a nuanced dialogue about dietary substitutions.

4.2. Closed-loop and continuous monitoring systems

While the aforementioned Mixture of Experts architectures brilliantly address the precision and accuracy of single-instance nutritional assessments, the ultimate clinical goal of precision nutrition is the seamless, long-term management of an individual's longitudinal dietary behavior. Research in Phase I and Phase II was almost exclusively focused on discrete "Dietary Snapshots"—isolated moments in time where a user actively chose to log a meal. Phase III, however, is definitively characterized by the ambitious construction of Time-Continuous Closed-Loop Systems.

This monumental leap in dietary assessment philosophy is most ambitiously embodied in the recent multi-agent system study by Xu (2026). Their research team refused to stop at merely identifying the contents of a single plate; instead, they engineered a comprehensive feedback loop system deeply inspired by the principles of Cybernetics and control theory. In this advanced framework, the artificial intelligence is not a singular entity but is explicitly designed as a collaborative collective of specialized agents. A designated "Perception Agent" is tasked exclusively with handling real-time intake data processing, constantly updating its semantic understanding of the food environment. Simultaneously, a separate "Planning Agent" operates in the background, continuously maintaining and adjusting the user's daily Nutritional Budget based on metabolic goals. Finally, an "Execution Agent" translates these calculations into actionable, personalized recommendations for the user's next meal.

The core technological innovation driving this architecture is its rigorous "State Management" protocol. The multi-agent system maintains a dynamically updated User State Vector, a complex mathematical representation that continuously tracks circulating blood glucose levels (if integrated with biosensors), total cumulative calories consumed, the exact remaining macronutrient budget, and the user's self-reported or inferred satiety feedback. Whenever the user consumes a food item, the system does not simply log the caloric data into a static database. Instead, the perception agent registers the intake, immediately triggering the planning agent to recalculate the entire remaining nutritional quota for the day. This instantly initiates a cascade where the execution agent dynamically adjusts the recommendation strategy for the subsequent meal, perhaps suggesting a high-fiber, low-glycemic dinner if the user's lunch caused an unexpected carbohydrate spike. This real-time "Meal-Level Feedback Loop" fundamentally transforms dietary management from a passive, retrospective recording exercise into an active, predictive navigational system, truly realizing the dynamic metabolic equilibrium promised by the concept of personalized nutrition.

However, to successfully achieve such unobtrusive, 24/7 closed-loop monitoring, the paradigm of human-computer interaction regarding data acquisition must undergo a radical evolution—transitioning from the high-friction requirement of "active photographing" to the frictionless paradigm of "passive sensing". Jiang et al. (2025) successfully demonstrated the necessary deep fusion of advanced hardware and algorithmic innovation required for this transition. Their research utilizes smart glasses equipped with miniaturized, low-power cameras, synergistically combined with cutting-edge Egocentric Vision technology, to achieve the fully passive capture of human dietary behaviors.

Unlike traditional smartphone applications that force users to deliberately interrupt their meals, pull out a device, and snap a photograph, this wearable system utilizes highly optimized background algorithms to automatically detect the specific biomechanical signatures of eating. By continuously monitoring the kinematics of hand-to-mouth movements and detecting the subtle cranial vibrations associated with chewing behaviors, the system autonomously triggers its cameras to capture high-resolution image streams only during active ingestion events. These continuous image streams are subsequently fed into robust backend large multimodal models for instantaneous nutritional analysis. Crucially, the system utilizes an integrated RAG module to generate immediate, context-aware health feedback based on the user's established medical profile. For example, the system is capable of deploying a micro-intervention—such as a gentle visual prompt popping up on the augmented reality display of the glasses—the exact moment a diabetic user reaches for a third piece of high-sugar dessert. This paradigm of "In-situ Monitoring" significantly reduces the psychological and physical compliance burden placed on the user. By completely automating the logging process, it directly solves the single most critical pain point in long-term epidemiological dietary tracking: user fatigue and subsequent study dropout. The emergence of this technology heralds the inevitable integration of dietary assessment into the invisible background of Ubiquitous Computing, transforming nutrition tracking into an "Always-on" physiological service entirely akin to modern continuous heart rate or sleep cycle monitoring.

4.3. Tool utilization in agentic systems

The most distinct and consequential technical characteristic defining Phase III is the acquisition of functional "Tool Use" capabilities by these multimodal agents. In the early, naive applications of LLMs in food science, researchers attempted to force models to solve all physical and geometric problems entirely via their internal neural network weights. This resulted in models attempting to statistically "hallucinate" the exact metric volume of a steak based purely on two-dimensional pixel patterns, a physically impossible task that led to massive estimation variances. However, the advanced literature emerging from 2025 and 2026 explicitly posits a new operational philosophy: Large Language Models should absolutely not be utilized as omnipotent, standalone calculators. Instead, they must be positioned as the Semantic Controllers of High-Level Reasoning, acting as cognitive orchestrators that learn to invoke external, highly deterministic algorithms to handle strict physical, geometric, and mathematical tasks.

This vital concept of "Tool-Augmented Learning" is particularly prominent in solving the persistent and notoriously difficult volume estimation challenge. Within the sophisticated architecture developed by Zhang et al. (2025), a profound shift in computational behavior occurs. When the LLM's visual encoder successfully identifies a known reference object within an image—such as a standard coin, a credit card, or a recognized plate diameter—the model actively suppresses its generative instinct. It no longer attempts to generate a highly flawed, natural language estimation of volume such as "the chicken breast appears to be about 200 milliliters." Instead, the LLM utilizes its code-generation capabilities to write and execute a precise Python Script in a sandboxed environment. This script explicitly calls upon traditional, mathematically rigorous Computer Vision libraries, such as OpenCV, to calculate exact pixel-to-millimeter ratios. Alternatively, if the hardware allows, the agent invokes specialized, deterministic Depth Estimation Models to construct accurate 3D spatial point clouds of the food matrix. In this advanced paradigm, the LLM's role shifts entirely to parsing the user's intent, orchestrating the sequence of tool calls, and interpreting the final numerical results returned by the scripts, while the heavy, error-prone geometric calculations are entirely outsourced to dedicated, infallible mathematical tools.

Similarly, this necessity for external tool invocation is heavily emphasized in the closed-loop state management system developed by Xu (2026). When their designated Planning Agent is tasked with generating a personalized meal plan, it does not fabricate dietary suggestions out of thin air using probabilistic text generation, which often leads to plans that fail to meet exact clinical micronutrient requirements. Instead, the agent invokes robust Linear Programming Solvers or sophisticated Convex Optimization Algorithms. The LLM acts as an intelligent translation layer; it parses the user's natural language dietary preferences, current metabolic state, and medical restrictions, translating these complex human factors into strict mathematical constraints and objective functions. It then hands these structured matrices over to the deterministic solver to compute the absolute optimal mathematical solution for the day's remaining nutritional budget. Once the solver returns the exact macronutrient distribution required, the LLM translates these cold mathematical results back into appetizing, culturally appropriate, natural language meal suggestions. This hybrid architecture provides a dual benefit: it absolutely guarantees the mathematical and clinical rigor of the medical advice—ensuring, for example, that a renal patient's phosphorus intake strictly meets nephrology targets—while simultaneously retaining the empathetic, conversational, and interactive flexibility of the LLM.

Ultimately, this profound capability for tool use marks a necessary regression and powerful fusion within the field of AI dietary assessment. It represents a deliberate move away from pure, unconstrained "Generative AI" and a triumphant return toward "Neuro-symbolic AI." This integrated approach rightfully acknowledges the deep neural network's unparalleled superiority in semantic intuition, visual perception, and natural language understanding, while simultaneously leveraging the undisputed strengths of traditional symbolic computation in strict logic, geometric precision, and mathematical rule-following. By empowering these multimodal agents with the ability to reach out and use specialized tools, the research community has gradually shattered the cognitive ceiling that previously limited LLMs in dealing with the geometric properties of the physical world and the strict numerical constraints of clinical health. This integration of semantic reasoning with deterministic tooling adds the critical piece to the puzzle, heading for the widespread deployment of fully automated, clinical-grade nutritional assessment systems.

5. Datasets and evaluation metrics

As Large Language Models (LLMs) evolve from generic natural language processing engines into specialized nutritional assessment tools, the field faces a severe "measurement crisis." Traditional Computer Vision (CV) evaluation paradigms are no longer sufficient to delineate the capability boundaries of Large Multimodal Models (LMMs). Consequently, the frontier literature from 2024 to 2026 witnesses a fundamental reconstruction of evaluation frameworks: transitioning from simple image classification benchmarks to comprehensive testbeds incorporating fine-grained ingredients, complex contexts, and rigorous statistical formulas.

5.1. Datasets

The first major leap in the evaluation paradigm is reflected in the shift of dataset construction, where researchers have begun to build multidimensional corpora specifically designed to test the nutritional reasoning capabilities of LLMs. Traditional datasets primarily focused on whether a model could correctly distinguish food categories, whereas next-generation datasets delve into the physical and chemical properties at the ingredient level.

To address the task conflicts and the lack of nutritional annotations faced during the fine-tuning of multimodal large models, Jiao et al. (2024) introduced the Uni-Food dataset. This platform integrates over 100,000 images and groundbreakingly provides categories, ingredients, recipes, and detailed ingredient-level nutritional information, supplying the necessary data breadth for training Mixture-of-Experts (MoE) models. Similarly, to capture subtle visual cues determining caloric density (such as the sheen of surface oil or the viscosity of sauces), Yao et al. (2024) curated CalData. This massive corpus containing 330,000 high-quality image-text pairs is specifically tailored for aligning visual features with nutritional descriptions. To tackle the complexity of highly processed foods, Qi et al. (2025) proposed the FastFood dataset, encompassing 84,446 images across 908 fast-food categories with exhaustive ingredient distribution labels, driving the deep fusion of visual and ingredient features. Furthermore, Romero-Tapiador et al. (2025a) constructed the FoodNExTDB dataset featuring 9,263 images rigorously labeled by nutrition experts. Notably, this dataset introduces annotations for 9 different cooking styles (e.g., "fried" or "boiled"), recognizing that the physical processes of cooking directly dictate the final macronutrient retention rates.

Beyond the expansion of visual data, the philosophy of dataset construction has extended toward the extremes of "pure cognitive reasoning" and "environmental context." The release of NutriBench by Dhaliwal et al. (2025) marks the field's transition from "visual perception testing" to "cognitive reasoning testing." NutriBench is a unique large-scale dataset containing 11,857 detailed natural language meal descriptions derived from real-world dietary intake data. The core insight behind this design is that in many authentic clinical scenarios (such as remote surveys in elderly care), image data is often missing or highly blurred. In these instances, the LLM must possess the capability—akin to a human dietitian—to infer nutritional composition solely from text descriptions (e.g., "a bowl of pumpkin soup with a swirl of full-fat cream"). Tests on this benchmark exposed the severe hallucination risks of generalist models when handling unstandardized text. In contrast, Coburn et al. (2025) addressed a long-neglected dimension by introducing the ACETADA benchmark. This dataset integrates rich "Contextual Metadata" for the first time, including eating timestamps and venue types derived from GPS coordinates (e.g., "fast-food outlet" vs. "home kitchen"). Research indicates that when models incorporate this metadata as a Bayesian Prior into their reasoning, their prediction accuracy for hidden fat and sodium content is significantly corrected. A qualified next-generation dataset cannot merely be an aggregation of pixels; it must include a metadata layer capable of triggering causal reasoning within the model.

5.2. Evaluation metrics

Accompanying the evolution of datasets, the mathematical metrics for measuring model performance have also undergone a profound transformation from "Computer Science-oriented" to "Nutritional Science-oriented." In initial phase studies, common metrics were Top-1 Accuracy or Intersection over Union (IoU), which are useful for evaluating object detection but lack practical significance for guiding clinical nutritional interventions. Recent literature establishes continuous error analysis, correlation distributions, and limits of clinical agreement as the core evaluation standards.

To endow evaluation results with intuitive physical meaning, researchers established the Mean Absolute Error (MAE) as the core error metric, calculated as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

where n is the total number of samples, y_i is the Ground Truth obtained via laboratory analysis or dietitian coding, and $\hat{y}_i$ is the model's predicted value. Compared to the Root Mean Square Error (RMSE), MAE is less sensitive to extreme outliers and more accurately reflects the average energy deviation users encounter during daily tracking (e.g., "the model's average absolute error is 50 kcal").

In the assessment of macronutrients (proteins, fats, carbohydrates), accurately capturing the distribution patterns of various nutrients within the overall diet is paramount. Therefore, the Pearson Correlation Coefficient (r) is widely utilized:

$$r = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2 \sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}}$$

where $\bar{y}$ and $\bar{\hat{y}}$ represent the sample means of the true values and the predicted values, respectively. This formula measures the linear agreement between the predicted trend and the true trend by calculating the ratio of the covariance to the product of the standard deviations. Even if the model's absolute values exhibit a systematic shift, a high correlation coefficient proves that the model comprehends the relative proportions of the dietary structure.

More critically, the medical statistical tool Bland-Altman Analysis has become the "touchstone" for verifying the clinical viability of AI models (as advocated by Fridolfsson et al. (2025)). This method establishes limits of agreement by calculating the difference $d_i = y_i - \hat{y}_i$ for each measurement. First, the Bias ($\bar{d}$) is calculated to measure systematic error:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$$

Subsequently, using the standard deviation of the differences (SD_d), the 95% Limits of Agreement (LoA) are calculated:

$$LoA = \bar{d} \pm 1.96 \times SD_d$$

The Bland-Altman plot graphs the differences d_i on the Y-axis against the averages $\frac{y_i+\hat{y}_i}{2}$ on the X-axis. Through this visualized mathematical model, researchers have revealed a dangerous "Proportional Bias" prevalent in current generalist models: models perform adequately on low-calorie foods, but as the volume or caloric density of the food increases, their tendency to underestimate amplifies linearly. This fine-grained error analysis, based on rigorous mathematical formulas, signifies that the evaluation system in the field of dietary assessment has officially transitioned from single technical scoring to clinical-grade data validation.

6. Challenges and Future Directions

Despite the exciting prospects demonstrated by Phase III agentic architectures, realizing clinical-grade automated dietary assessment on a global scale requires crossing three monumental barriers: the physical bottleneck of monocular vision, the ontological bias of global diets, and the privacy paradox of continuous monitoring.

6.1. The Volumetric Bottleneck

Even equipped with the most advanced reasoning agents, current vision models remain constrained by a fundamental optical physical limit known as depth ambiguity. The empirical data provided by Fridolfsson et al. (2025) not only revealed specific estimation errors but fundamentally touched upon this physical essence. A single RGB image is inherently a mathematical projection of a three-dimensional world onto a two-dimensional plane, a process that inevitably loses absolute scale and depth information. While human dietitians can infer volume through prior knowledge and spatial intuition, this inference quickly fails for non-standardized foods, such as a complex mound of mashed potatoes or a bowl of irregular stew.

Further analysis of the data demonstrated that leading multimodal models often exceed 40% error rates when estimating high-piled or high-density foods. This occurs because models rely primarily on the segmentation area as a proxy variable for volume, completely failing to perceive the hidden "thickness" or "density" within the food matrix. Although Tanabe and Yanai (2025) attempted to mitigate this via Chain-of-Thought (CoT) reasoning—for example, prompting the model to explicitly deduce that a deep bowl implies more food underneath—such reasoning often degenerates into ungrounded guessing without clear visual cues. If the depth information provided by the visual encoder is inherently flawed, even the most rigorous downstream logical reasoning cannot yield correct calorie values, resulting in a classic "garbage in, garbage out" scenario. Future breakthroughs will likely not emerge from simply scaling up the parameters of language models, but rather from the integration of multi-view geometry with multimodal models, or the utilization of mobile LiDAR sensors to acquire real depth maps. This paradigm shift would restore volume estimation from an experience-based textual reasoning problem back to a deterministic, geometry-based computational problem.

6.2. Long-tail Foods and Global Adaptability

Current large multimodal models exhibit severe Western-centrism, which is not merely a problem of imbalanced datasets but a profound misalignment at the ontological level. Carrillo-Larco et al. (2025) provided a compelling counter-example in their evaluation of non-Western diets. When faced with traditional Peruvian lunches, the models not only failed to recognize certain indigenous ingredients but, more critically, failed to understand the local "invisible food matrix." For example, in many non-Western dietary cultures, oils, sugars, or starches are not merely superficial toppings but are deeply fused into the complex cooking process, such as cornstarch used for thickening or animal fats used as a foundational flavor base.

Because mainstream pre-training corpora are heavily dominated by Euro-American recipes, these models inevitably acquire an inherently Western culinary logic (e.g., assuming salads are always raw and meat is typically grilled). When this logic is forcibly applied to the complex stewing, frying, or mixed cooking techniques prevalent in Asian, African, or Latin American cuisines, the models systematically underestimate fat and carbohydrate contents. This presents a classic long-tail distribution challenge: for highly standardized items like burgers and pizzas representing the head of the data, the models perform nearly perfectly; however, for the thousands of regional dishes consumed daily by billions of people representing the tail, the models are virtually blind. Solving this challenge cannot rely solely on superficial parameter fine-tuning. It requires interdisciplinary efforts to construct culturally aware knowledge graphs, enabling the model to truly understand ingredient substitution relationships and complex culinary physics across diverse cultural contexts, thereby achieving true global adaptability and health equity in dietary assessment.

6.3. Edge AI and Privacy

As dietary assessment inevitably evolves towards continuous passive monitoring, beautifully demonstrated by the wearable smart glass systems developed by Jiang et al. (2025), data privacy has emerged as the core ethical barrier between technological innovation and real-world deployment. To achieve high-precision nutritional analysis, current architectures typically require continuously uploading high-definition images or live video streams to massive cloud-based models for processing. This means users' every meal, including their private social environments and background activities during eating, is constantly exposed to cloud servers. Such a highly intrusive architecture poses a fatal blow to user compliance and widespread clinical adoption.

Researchers have deeply explored this dilemma and proposed the critical necessity of transitioning to Edge AI. The future trajectory involves running Small Language Models (SLMs) or specialized models optimized via knowledge distillation directly on end-side devices, such as smart glasses, watches, or mobile phones. These localized models do not require the generic generalization capabilities of foundation models—like writing poetry or generating code—but only need to focus their computational power exclusively on classifying food items and estimating portions. Through advanced quantization and model pruning technologies, systems can perform the preliminary processing of sensitive data directly on local silicon. For instance, the edge model can extract only abstract food feature vectors for upload rather than transmitting raw images, thereby ensuring "data residency" and drastically mitigating privacy risks. However, this introduces a new technical contradiction: how to maintain inference precision comparable to hundred-billion-parameter cloud models while operating under the strictly limited compute budgets and battery lifespans of mobile devices. This tension between computational power and precision represents the core trade-off problem that future hardware architects and nutritional algorithm engineers must collaboratively conquer.

7. Conclusion

The profound integration of Large Language Models (LLMs) and Multimodal Agents marks a definitive paradigm shift in the field of nutritional assessment, successfully dismantling two historical bottlenecks simultaneously: the cognitive fragility of human memory and the semantic superficiality of traditional machine learning (ML). For decades, nutritional epidemiology has been built on fragile foundations—namely, subjects' subjective retrospective accounts of their dietary behavior, such as 24-hour recalls (24HR) and Food Frequency Questionnaires (FFQ). This "recall-based" paradigm intrinsically introduced systematic biases that obstructed the true implementation of precision nutrition. Subsequently, while the first wave of digitization employed traditional Computer Vision (CV) and ML algorithms to automate food logging, these legacy models remained trapped in a reductive "pixel-to-label" classification paradigm. They were highly capable of classifying an image as a "burger" or a "salad," but were fundamentally blind to hidden ingredients, culinary logic, and 3D volumetric reasoning. Traditional ML lacked the cognitive loop required to translate visual features into rigorous nutritional computations.

The literature from 2024 to 2026 reviewed in this article indicates that we are undergoing a profound epistemological revolution. The trajectory of technological evolution is clearly visible:

Phase I utilized the perception-enhanced capabilities of generalist models to solve the "Recognition" problem ("What is it?"), achieving zero-shot identification of standardized foods.

Phase II introduced architecturally augmented capabilities of modular LLM pipelines—integrating RAG, fine-tuning, and Chain-of-Thought—to solve the "Composition" problem ("What is in it?"), connecting AI with authoritative nutritional science.

Phase III is marching towards a new reasoning-driven era through agentic architectures and tool use, solving the "Action" problem ("How to manage?"), realizing continuous, closed-loop, and clinical-grade dietary automation.

This evolution is not merely an improvement in accuracy but a fundamental shift in the role of AI: from a passive recording tool to an active "Digital Dietitian" capable of causal reasoning. Although the physical bottlenecks of volume estimation and challenges of cultural adaptability persist, with the maturation of Mixture of Experts (MoE) models and Edge Intelligence, automated dietary assessment is gradually approaching the critical point of clinical utility. Future research should no longer obsess over whether a model can recognize an apple, but should dedicate itself to constructing symbiotic intelligent systems that understand the complexity of human dietary behavior, respect cultural diversity, and protect individual privacy. We have reason to believe that this technological leap will finally unravel the complex causal chains between diet and health, bringing unprecedented precision intervention capabilities to global public health.

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